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CIRCULAR SUTURE CONSTRUCTS AND METHODS FOR USE

Abstract

A suture construct includes, inter alia, a sheath and a flexible strand threaded through the sheath to configure the sheath in a circular or oblong shape. The suture construct may be utilized in various tissue reconstruction procedures. The suture construct can be connected to a second suture construct to form a surgical assembly.

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Background/Summary

RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/942,492 filed Sep. 12, 2022; which is a continuation of U.S. patent application Ser. No. 16/406,428 filed May 8, 2019, now U.S. Pat. No. 11,452,518 granted Sep. 27, 2022; which is a continuation of U.S. patent application Ser. No. 15/277,461 filed Sep. 27, 2016, now U.S. Pat. No. 10,307,154 granted Jun. 4, 2019; the entirety of which are herein incorporated by reference.

BACKGROUND

[0002] This disclosure relates to suture constructs and methods of using the suture constructs in various tissue reconstruction procedures.

[0003] Orthopedic procedures are often performed to repair musculoskeletal injuries. For example, soft tissue may tear away from bone during vigorous exercise or sporting activities. When tears occur, reattachment is often necessary to repair the damaged tissue. Suture anchors and buttons are two types of surgical devices that have been developed to facilitate these repairs; however, additional advancements in this field of technology are desired.

SUMMARY

[0004] This disclosure relates to suture constructs and methods of use. A suture construct may include a sheath and a flexible strand threaded through the sheath. Together, the sheath and the flexible strand form a “soft” suture construct. The suture constructs can be used in various tissue reconstruction procedures, including but not limited to, ACL/PCL reconstructions and meniscal tear repairs.

[0005] A suture construct according to an exemplary aspect of this disclosure includes, inter alia, a sheath and a flexible strand threaded through the sheath to establish a bridge configuration that includes at least two strand portions that cross over one another outside of the sheath.

[0006] A method according to another exemplary aspect of this disclosure includes, inter alia, utilizing a suture construct in a tissue reconstruction procedure. The suture construct includes a sheath and a flexible strand threaded through the sheath to configure the sheath in a circular or oblong shape.

[0007] A surgical assembly according to yet another exemplary aspect of this disclosure includes, inter alia, a first suture construct, a second suture construct, and a flexible strand that connects the first suture construct to the second suture construct. The flexible strand includes a sliding knot.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 illustrates a suture construct according to a first embodiment of this disclosure.

[0009] FIG. 2 is a cross-sectional view of the suture construct of FIG. 1.

[0010] FIG. 3 schematically illustrates a method of constructing the suture construct of FIGS. 1 and 2.

[0011] FIG. 4 illustrates a suture construct according to a second embodiment of this disclosure.

[0012] FIG. 5 schematically illustrates a method of constructing the suture construct of FIG. 4.

[0013] FIG. 6 schematically illustrates a method of using a suture construct as part of a tissue reconstruction procedure.

[0014] FIG. 7 illustrates a surgical assembly that includes a first suture construct and a second suture construct.

[0015] FIGS. 8A-8F schematically illustrate a method of constructing the surgical assembly of FIG. 7.

[0016] FIG. 9 schematically illustrates a method of using a surgical assembly of suture constructs as part of a tissue reconstruction procedure.

DETAILED DESCRIPTION

[0017] This disclosure relates to suture constructs and their related methods of use. A suture construct can include a sheath and a flexible strand threaded through the sheath to form a “soft” suture construct. The suture constructs can be used in various tissue reconstruction procedures, including but not limited to, ACL/PCL reconstructions and meniscal tear repairs.

[0018] A suture construct according to an exemplary aspect of this disclosure includes, inter alia, a sheath and a flexible strand threaded through the sheath to establish a bridge configuration that includes at least two strand portions that cross over one another outside of the sheath.

[0019] In a further embodiment, a sheath includes a tubular body that extends between a first open end and a second open end.

[0020] In a further embodiment, a second open end of a sheath is adjacent to and faces a first open end when a sheath is configured in a circular or oblong shape.

[0021] In a further embodiment, a flexible strand of a suture construct exits through a sheath at multiple splice points.

[0022] In a further embodiment, a flexible strand establishes a bridge configuration inside a perimeter of a circular or oblong shape of a sheath of a suture construct.

[0023] In a further embodiment, a bridge configuration is a double bridge configuration.

[0024] In a further embodiment, a double bridge configuration includes multiple strand portions of a flexible strand exiting through multiple splice points of a sheath.

[0025] In a further embodiment, a second suture construct is connected to a suture construct.

[0026] In a further embodiment, a flexible strand of a suture construct or a second flexible strand of a second suture construct connects the suture construct and the second suture construct.

[0027] In a further embodiment, a sliding knot is formed in a flexible strand.

[0028] In a further embodiment, a flexible strand is threaded through opposing ends and at least four splice points of a sheath.

[0029] In a further embodiment, a sheath and a flexible strand are both made exclusively of soft, suture based materials.

[0030] In a further embodiment, a flexible strand includes a first strand portion that extends outside of a sheath and connects between opposing splice points of the sheath.

[0031] In a further embodiment, a second strand portion crosses over a first strand portion to establish an X-shaped bridge configuration.

[0032] A method according to another exemplary aspect of this disclosure includes, inter alia, utilizing a suture construct in a tissue reconstruction procedure. The suture construct includes a sheath and a flexible strand threaded through the sheath to configure the sheath in a circular or oblong shape.

[0033] In a further embodiment, a sheath is configured in a circular shape prior to utilizing a suture construct in a tissue reconstruction procedure.

[0034] In a further embodiment, a tissue reconstruction procedure is an ACL reconstruction procedure.

[0035] In a further embodiment, the ACL reconstruction procedure includes attaching a flexible strand to a graft, and suspending the graft within a bone tunnel with a suture construct.

[0036] In a further embodiment, a tissue reconstruction procedure is a meniscal tear repair procedure.

[0037] A surgical assembly according to yet another exemplary aspect of this disclosure includes,

inter alia, a first suture construct, a second suture construct, and a flexible strand that connects the first suture construct to the second suture construct. The flexible strand includes a sliding knot. [0038] FIGS. **1** and **2** illustrate a suture construct **10**. The suture construct **10** is configured for use in various tissue reconstruction procedures and may be fixated within a joint to repair torn tissue. ACL/PCL reconstructions and meniscal tear repairs are but two non-limiting examples of tissue reconstruction procedures which could benefit from the suture constructs of this disclosure.

[0039] The suture construct **10** may be referred to as a “soft” construct because it is formed of soft materials such as yarns, fibers, filaments, strings, fibrils, strands, sutures, etc., or any combination of such materials. The soft materials may be synthetic or natural materials, or combinations of synthetic and nature materials, and may be biodegradable or non-biodegradable within the scope of this disclosure. In an embodiment, the suture construct **10** is made exclusively of soft, suture based materials. The soft materials confer the ability to be inserted into or through tissue (e.g., bone, ligament, tendon, etc.) and then bunch together, collapse, expand and/or change shape to fixate the suture construct **10** relative to the tissue.

[0040] The exemplary suture construct **10** may include a sheath **12** and a flexible strand **14** that is threaded through the sheath **12** in a manner that configures the sheath **12** in a circular or oblong shape. The sheath **12** is configured in the circular or oblong shape in its non-deployed state. Stated another way, the circular or oblong shape is formed before using the suture construct **10** in a tissue reconstruction procedure, or before the suture construct **10** has been bunched together (i.e., deployed) by tensioning the flexible strand **14**.

[0041] The sheath **12** includes a tubular body **16** that extends between opposing ends **18A**, **18B**. The opposing ends **18A**, **18B** may be open ends. The tubular body **16** established a bore **20** that extends between the opposing ends **18A**, **18B** for accommodating the flexible strand **14**. When positioned in the circular or oblong shape, the opposing ends **18A**, **18B** of the sheath **12** are adjacent and may be parallel to one another.

[0042] In an embodiment, the sheath **12** is a tubular sleeve made of a flexible material, such as a braided, woven, or knitted structure made of yarns, fibers, filaments, sutures, or other similar materials, or combinations of these materials. In another embodiment, the sheath **12** is constructed of polyester suture material. However, other flexible materials may also be suitable for constructing the sheath **12**.

[0043] The flexible strand **14** is passed through the bore **20** of the sheath **12**. The flexible strand **14** can be threaded through the bore **20** in one or more passes to configure the sheath **12** in a circular or oblong shape in its non-deployed state. The flexible strand **14** may also be tensioned to assist in deploying the sheath **12** after inserting the suture construct **10** into or through tissue.

[0044] In another embodiment, the flexible strand **14** may exit the tubular body **16** through a plurality of splice points of the tubular body **16**. The splice points are spaced from the opposing ends **18A**, **18B** of the sheath **12**. The flexible strand **14** may include one or more strand portions **25** that extend outside of the sheath **12** and connect between opposing splice points. In an embodiment, the strand portions **25** establish a bridge configuration **22** within the center space/opening that is formed by configuring the sheath **12** in the circular or oblong shape. The flexible strand **14** may include any length and also includes free ends **24** that extend from the bridge configuration **22** outside of the bore **20** of the sheath **12**. In an embodiment, at least a portion of the strand portions **25** cross over one another to establish an X-shaped bridge configuration **22**.

[0045] In an embodiment, the flexible strand **14** is a suture. Non-limiting examples of suitable sutures include FiberWire®, TigerWire®, or FiberChain® suture, which are each available from Arthrex, Inc., although any type of suture may be utilized, including cored or coreless sutures. In another embodiment, the flexible strand **14** is flat suture, such as FiberTape® or SutureTape®, which are also available from Arthrex, Inc. The flexible strand **14** could include any soft, flexible strand of material.

[0046] FIG. 3, with continued reference to FIGS. 1-2, illustrates a method of constructing the suture construct **10**. In addition to threading the flexible strand **14** through the opposing ends **18A**, **18B** of the sheath **12**, the flexible strand **14** may be threaded through a plurality of splice points **SP1**, **SP2**, **SP3**, and **SP4**. The splice points **SP1**, **SP2**, **SP3**, and **SP4** are located in sequential order moving in a clockwise direction from the opposing end **18A** toward the opposing end **18B** of the sheath **12**.

[0047] An exemplary spacing between the splice points **SP1**, **SP2**, **SP3**, and **SP4** and between the opposing ends **18A**, **18B** and the splice points **SP1** and **SP4**, respectively, is shown in FIG. 3. To better illustrate the exemplary spacing, the splice points **SP1**, **SP2**, **SP3**, and **SP4** are depicted along a linear axis **A** in addition to their depiction relative to the circular or oblong shaped suture construct **10**. The linear axis **A** represents a longitudinal axis of the sheath **12** in its straightened configuration (i.e., before configuring the sheath **12** in the circular or oblong shape). In an embodiment, the splice point **SP1** and the splice point **SP4** are spaced a first distance **D1** from the opposing end **18A** and the opposing end **18B**, respectively. In another embodiment, a second distance **D2** is provided between splice points **SP1** and **SP2**, between splice points **SP2** and **SP3**, and between splice points **SP3** and **SP4**. The second distance **D2** may be larger than the first distance **D1**, in yet another embodiment. The actual spacing between the splice points **SP1**, **SP2**, **SP3**, and **SP4** may vary depending on the size of the sheath **12** and the type of reconstruction procedure the suture construct **10** is being used for, among other design criteria.

[0048] In an embodiment, the flexible strand **14** is threaded through the sheath **12** as follows in order to configure the suture construct **10** in the circular or oblong shape: [0049] Step 1: Insert a first free end of the free ends **24** of the flexible strand **14** through the opposing end **18B** of the sheath **12** and then move the flexible strand **14** in a counterclockwise direction through the bore **20** of the sheath **12**. Exit the flexible strand **14** from the sheath **12** through the splice point **SP1**. [0050] Step 2: Insert a second free end of the free ends **24** of the flexible strand **14** through the opposing end **18A** of the sheath **12** and then move the flexible strand **14** in a clockwise direction through the bore **20** of the sheath **12**. Exit the flexible strand **14** from the sheath **12** through the splice point **SP2**. [0051] Step 3: Insert the strand portion **25** of the flexible strand **14** exiting splice point **SP2** through the splice point **SP4** and reenter the bore **20** of the sheath **12**. Next, move the flexible strand **14** in a clockwise direction until it exits through the opposing end **18B**. [0052] Step 4: Insert the flexible strand **14** back through the opposing end **18A** and then exit the flexible strand **14** through the splice point **SP3**. [0053] Step 5: Tension the free ends **24** of the flexible strand **14** that exit from the splice points **SP1** and **SP3** to configure the sheath **12** in the circular or oblong shape. The opposing ends **18A**, **18B** are directly adjacent one another and face one another when the sheath **12** is positioned in the circular or oblong shape. [0054] Step 6: Route the free ends **24** of the flexible strand **14** that exit from the splice points **SP1** and **SP3** around the strand portion **25** of the flexible strand **14** extending between the splice points **SP2** and **SP4** to establish an “X” shaped bridge configuration **22** inside a perimeter of the circular or oblong shaped sheath **12**. [0055] Step 7: Splice the free ends **24** of the flexible strand **14** together, such as by tying the free ends **24** together using a knot. If necessary, the free ends **24** of the flexible strand **14** can be tipped with a suitable material to avoid fraying.

[0056] FIG. 4 illustrates another exemplary embodiment of a suture construct **110**. In this disclosure, like reference numerals designate like elements where appropriate and reference numerals with the addition of 100 or multiples thereof designate modified elements that are understood to incorporate the same features and benefits of the corresponding original elements. The suture construct **110** includes a sheath **12** and a flexible strand **14** threaded through the sheath **12**. The sheath **12** is shown as slightly transparent in FIG. 4 to better illustrate how the flexible strand **14** is threaded through the sheath **12**.

[0057] The flexible strand **14** is threaded through the sheath **12** to configure the sheath **12** in a circular or oblong shape. The suture construct **110** of FIG. 4 is similar to the suture construct **10** of

FIGS. 1-3. However, in this embodiment, the flexible strand **14** is threaded through the sheath **12** to establish a double bridge configuration **122** in the center of the circular or oblong shaped sheath **12**. In the double bridge configuration **122**, two strand portions **25** of the flexible strand **14** exit the sheath **12** through each splice point SP1, SP2, SP3, and SP4.

[0058] FIG. 5, with continued reference to FIG. 4, schematically illustrates a manner in which the flexible strand **14** can be threaded through the sheath **12** to construct the suture construct **110** that includes the double bridge configuration **122**. In an embodiment, the flexible strand **14** is threaded through the sheath **12** as follows to construct the suture construct **110**: [0059] Step 1: Insert a first free end of the free ends **24** of the flexible strand **14** through the splice point SP1 and then move the flexible strand **14** in a counterclockwise direction through a bore **20** of the sheath **12**. Exit the flexible strand **14** from the sheath **12** through the splice point SP2. [0060] Step 2: Insert the strand portion **25** of the flexible strand **14** exiting the splice point SP2 through the splice point SP4 and then exit the flexible strand **14** from the sheath **12** through the splice point SP3. [0061] Step 3: Insert the strand portion **25** of the flexible strand **14** exiting the splice point SP3 back through the splice point SP1 and then exit the flexible strand **14** from the sheath **12** through the opposing end **18A**. [0062] Step 4: Insert the flexible strand **14** through the opposing end **18B** and then exit the flexible strand **14** back through the splice point SP4. [0063] Step 5: Insert the strand portion **25** of the flexible strand **14** exiting the splice point SP4 back through the splice point SP2 and then exit the flexible strand **14** back through the splice point SP3. [0064] Step 6: Tension the free ends **24** of the flexible strand **14** that exit from the splice points SP1 and SP3 to configure the sheath **12** in the circular or oblong shape. The opposing ends **18A**, **18B** are directly adjacent one another and face one another when the sheath **12** is positioned in the circular shape. [0065] Step 6: Route the free ends **24** of the flexible strand **14** that exit from the splice points SP1 and SP3 around the strand portions **25** of the flexible strand **14** extending between the splice points SP2 and SP4 to establish an “X” shaped double bridge configuration **122** inside a perimeter of the circular or oblong shaped sheath **12**. [0066] Step 7: Splice the free ends **24** of the flexible strand **14** together, such as by tying or stitching the free ends **24** of the flexible strand **14** together.

[0067] FIG. 6 illustrates an exemplary surgical use of the suture construct **10** (or suture construct **110**) during a tissue reconstruction procedure, such as an ACL reconstruction procedure. However, it should be understood that this disclosure is not limited to ACL reconstruction procedures, and the suture constructs of this disclosure could be used in a variety of tissue repair or reconstruction procedures within the scope of this disclosure.

[0068] The suture construct **10**, **110** may be implanted within a joint **26** (e.g., a knee joint) to repair a torn tissue (e.g., a torn ACL). Prior to positioning the suture construct **10** within the joint **26**, a bone tunnel **28** (i.e., a socket, hole, opening, etc.) is formed in a bone **30** (e.g., a femur). The bone tunnel **28** may be formed using known drilling techniques to establish voids within the bone **30** for accommodating the suture construct **10**, **110**. In an embodiment, the sheath **12** of the suture construct **10**, **110** is already configured in a circular or oblong shape prior to implantation of the suture construct **10** within the joint **26**. Thus, the suture construct **10**, **110** is configured in the circular or oblong shape in what is referred to as its pre-deployed state. In another embodiment, the suture construct **10**, **110** is deployed into the joint **26** in a straight configuration (e.g., the sheath **12** is generally straight along linear axis A) and can then be tensioned into the circular or oblong shape by pulling on the free ends **24**.

[0069] In an exemplary embodiment, the suture construct **10** is implanted by passing the suture construct **10**, **110** through the bone tunnel **28**. The suture construct **10**, **110** may be pushed or pulled through the bone tunnel **28**.

[0070] The free ends **24** of the flexible strand **14** of the suture construct **10**, **110** may be attached to a graft **32**. Therefore, once the suture construct **10**, **110** is implanted within the joint **26**, the graft **32** is suspended within the bone tunnel **28** to reconstruct or repair the ACL. Fixation of the graft **32** to a second bone (not shown), such as a tibia, can be achieved using known techniques. The graft **32**

may be tensioned to tension the flexible strand **14** and thereby cause the suture construct **10**, **110** to bunch up and thus, collapse, expand, and/or change shape to fixate relative to the bone tunnel **28** of the bone **30**.

[0071] FIG. **7** illustrates a surgical assembly **34** that includes a first suture construct **10A** and a second suture construct **10B**. Although depicted including suture constructs similar to those of FIGS. **1-3**, the surgical assembly **34** could also be made up of suture constructs **110** of FIGS. **4-5**, or a combination of the suture construct **10** of FIGS. **1-3** and the suture construct **110** of FIGS. **4-5**.

[0072] Each suture construct **10A**, **10B** includes a sheath **12** and a flexible strand **14**. The flexible strands **14** are threaded through the sheaths **12** to configure the sheaths **12** in a circular or oblong shape. In an embodiment, the first suture construct **10A** is connected to the second suture construct **10B** by a flexible strand, such as the flexible strand **14** of the second suture construct **10B**. For example, the free end **24** of the flexible strand **14** of the second suture construct **10B** may be looped over the bridge configuration **22** of the first suture construct **10A** to connect the first and second suture constructs **10A**, **10B**.

[0073] The flexible strand **14** that connects the first and second suture constructs **10A**, **10B** may include a sliding knot **36**. The sliding knot **36** can be adjusted to alter the amount of slack between the first suture construct **10A** and the second suture construct **10B** and thus close a tissue tear, as discussed in greater detail below.

[0074] FIGS. **8A-8F** schematically illustrate a method for constructing the surgical assembly **34** of FIG. **7**. Referring first to FIG. **8A**, the free end **24** of the flexible strand **14** of the second suture construct **10B** is looped over the bridge configuration **22** of the first suture construct **10A**. The flexible strand **14** of the second suture construct **10B** may be longer than the flexible strand **14** of the first suture construct **10A** in order to provide a sufficient length to use the flexible strand **14** of the second suture construct **10B** for connecting the first and second suture constructs **10A**, **10B**. Looping the flexible strand **14** of the second suture construct **10B** in this manner creates a right tail lead portion **40** and a post portion **42** in the flexible strand **14** of the second suture construct **10B**.

[0075] Next, the right tail lead portion **40** is wrapped under the post portion **42** (see FIG. **8B**). The second suture construct **10B** is then wrapped over the post portion **42**, through an opening **44** between the post portion **42** and the right tail lead portion **40**, and then under the right tail lead portion **40** (see FIG. **8C**). The second suture construct **10B** is then wrapped around both the right tail lead portion **40** and the post portion **42** (see FIG. **8D**).

[0076] As shown in FIG. **8E**, the second suture construct **10B** is next wrapped under the post portion **42** back toward the opening **44** and over a crossed lead **46**. Finally, as shown in FIG. **8F**, the second suture construct **10B** is placed through the opening **44** and back under the right tail lead portion **40**. The second suture construct **10B** may be further tensioned in the direction **T1** to fully form the sliding knot **36** shown in FIG. **7**.

[0077] FIG. **9**, with continued reference to FIGS. **7** and **8A-8F**, illustrates an exemplary surgical use of the surgical assembly **34** during a tissue reconstruction procedure, such as a meniscal tear repair procedure. For example, the surgical assembly **34** may be inserted through a meniscus **50** to repair a meniscus tear **52**. The first suture construct **10A** is inserted through the meniscus **50** first, and then the second suture construct **10B** is inserted through the meniscus **50** at an adjacent location. Once inserted, the flexible strand **14** that connects the first and second suture constructs **10A**, **10B** is positioned on an opposite side of the meniscus **50** from the first and second suture constructs **10A**, **10B**.

[0078] The sliding knot **36** formed in the flexible strand **14** may then be moved to tension the flexible strand **14** and close the tear **52**. In an embodiment, a knot pusher is used to advance the sliding knot **36**. As the sliding knot **36** is advanced and tension accumulates in the flexible strand **14**, the first and second suture constructs **10A**, **10B** bunch together to form anchoring clusters **60**. The anchoring clusters **60** promote fixation of the surgical assembly **34** relative to the meniscus **50** and facilitate closing of the tear **52**.

[0079] The suture constructs of this disclosure provide increased bulkiness and rigidity and therefore exhibit increased pull-out strength compared to prior art constructs. The circular or oblong shape of the suture constructs of this disclosure maximizes the amount of construct retention surface area achieved during construct deployment.

[0080] Although the different non-limiting embodiments are illustrated as having specific components or steps, the embodiments of this disclosure are not limited to those particular combinations. It is possible to use some of the components or features from any of the non-limiting embodiments in combination with features or components from any of the other non-limiting embodiments.

[0081] It should be understood that like reference numerals identify corresponding or similar elements throughout the several drawings. It should be understood that although a particular component arrangement is disclosed and illustrated in these exemplary embodiments, other arrangements could also benefit from the teachings of this disclosure.

[0082] The foregoing description shall be interpreted as illustrative and not in any limiting sense. A worker of ordinary skill in the art would understand that certain modifications could come within the scope of this disclosure. For these reasons, the following claims should be studied to determine the true scope and content of this disclosure.

Claims

1. A suture construct, comprising: a sheath including a tubular body having an internal passageway, wherein the tubular body is configurable in a round shape to form a center space defined by a perimeter of the tubular body; a first flexible strand threaded through the internal passageway of the sheath, wherein the first flexible strand exits the internal passageway through the sheath at a plurality of splice points, the splice points include first and third splice points on opposite sides of the center space and second and fourth splices points on opposite sides of the center space, and the first flexible strand includes a first segment that extends along the internal passageway between the second splice point and the third splice point; and a second flexible strand looped over adjacent segments of the first flexible strand within the center space.
2. The suture construct as recited in claim 1, wherein the first flexible strand includes first and second free ends that are outside of the sheath.
3. The suture construct as recited in claim 1, wherein the round shape is circular.
4. The suture construct as recited in claim 1, wherein the tubular body is configurable in the round shape in response to tensioning the first flexible strand.
5. The suture construct as recited in claim 4, wherein the round shape of the tubular body is collapsable in response to tensioning the first flexible strand.
6. The suture construct as recited in claim 1, wherein the sheath and the first flexible strand are made of soft, suture-based materials.
7. The suture construct as recited in claim 1, wherein the tubular body extends between a first open end and a second open end, and the second open end is adjacent to the first open end when the tubular body is configured in the round shape.
8. The suture construct as recited in claim 1, wherein a second suture construct comprises the second flexible strand.
9. The suture construct as recited in claim 8, further comprising a sliding knot that connects the first and second flexible strands to each other.
10. The suture construct as recited in claim 1, wherein the first flexible strand is attached to a surgical device securable in bone.
11. The suture construct as recited in claim 10, wherein the surgical device is a graft.
12. A suture construct, comprising: a sheath including a tubular body having an internal passageway, wherein the tubular body is configurable from a straight shape to a round shape to

form a center space in response to tensioning the sheath; a first flexible strand threaded through the internal passageway, wherein the first flexible strand exits the internal passageway through the sheath at a plurality of splice points, the plurality of splice points include at least four splice points distributed about a perimeter of the round shape, the first flexible strand includes a first segment that extends between an adjacent pair of the splice points, and the first segment exits from the internal passageway into the center space at the adjacent pair of the splice points; and a second flexible strand having a free end looped over the first flexible strand within the center space.

13. The suture construct as recited in claim 12, wherein the round shape is circular.

14. The suture construct as recited in claim 12, wherein the sheath and the first flexible strand are made of soft, suture-based materials.

15. The suture construct as recited in claim 12, wherein the first flexible strand includes first and second free ends that are outside of the sheath.

16. The suture construct as recited in claim 15, wherein the round shape of the tubular body is collapsible in response to tensioning the first and second free ends of the first flexible strand.

17. The suture construct as recited in claim 12, wherein the tubular body extends between a first open end and a second open end, the second open end is adjacent to and faces the first open end when the tubular body is configured in the round shape, and the internal passageway extends between the first and second open ends.

18. The suture construct as recited in claim 12, further comprising a sliding knot formed in the first flexible strand or the second flexible strand to connect the first and second flexible strands to each other.

19. The suture construct as recited in claim 12, wherein the first flexible strand is attached to a surgical device securable in bone.

20. The suture construct as recited in claim 19, wherein the surgical device is a graft adapted for suspension within a bone tunnel.
